

Specification part B
for Transfer Mechanism System

Plant	ORION
Supplier:	Open
Project name:	Transfer Mechanism System
Country specification part D:	Orion
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1. Overview

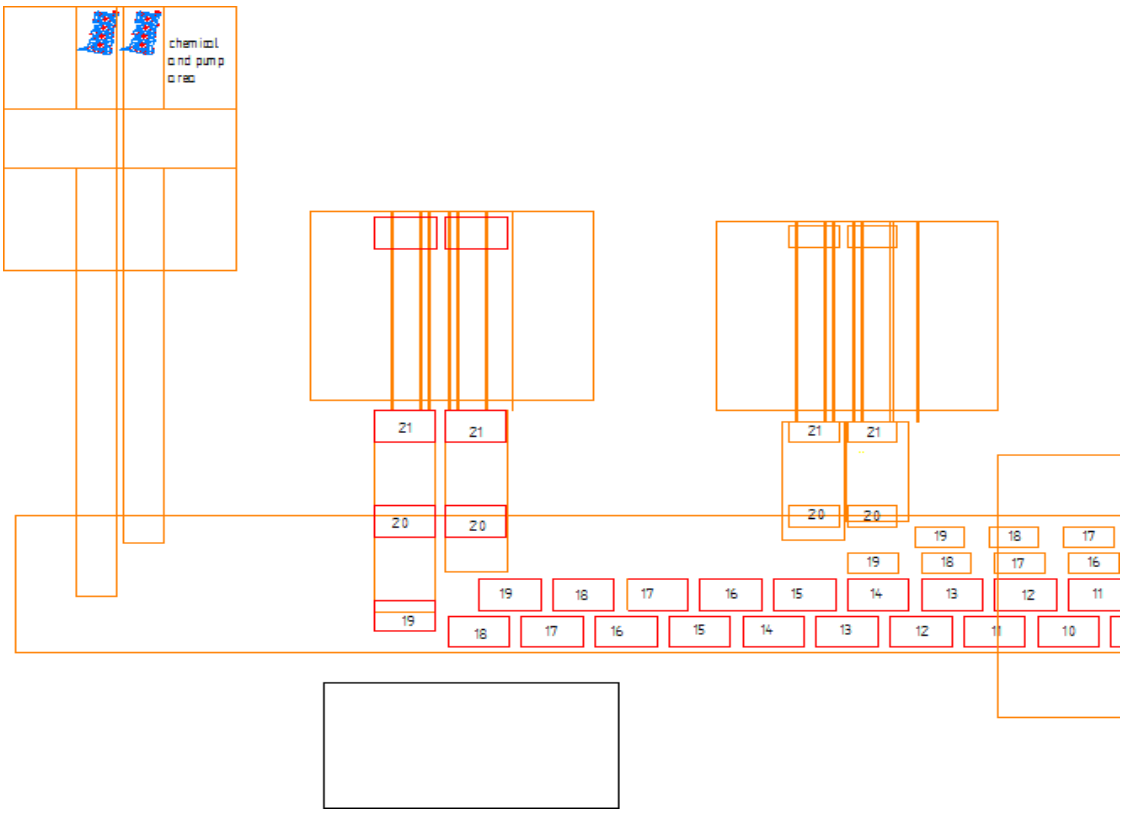
Scope

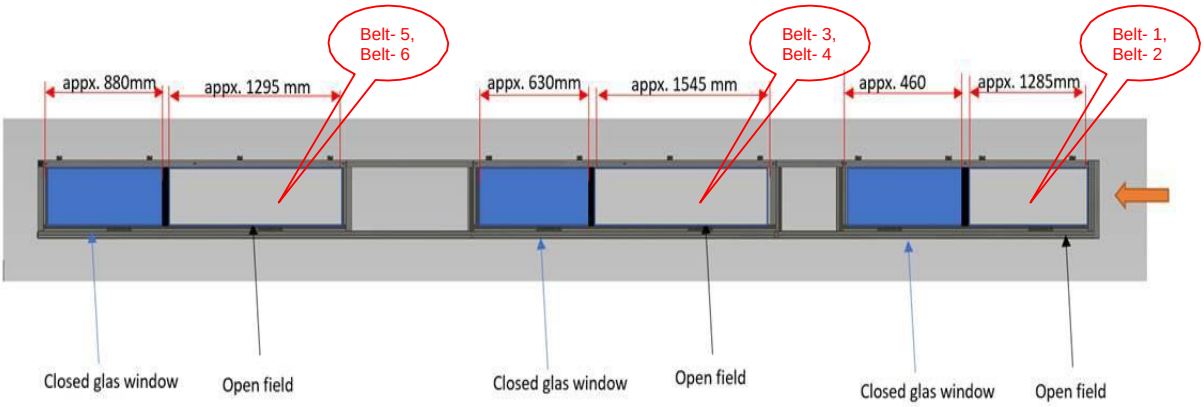
Transferring of Coated Layers from the Applicator machines to the feed zone of the ELPO curing oven.

General Information

- Screen Printing Machine(SPT)
  - We have 2 Screen Printed machines (SPT) with double cavities. Each cavities has output rate of one layer at a time, i.e. 2 layers output per stroke of the screen print machine. Each cavities have 2 belts on which layers are pushed out of the screen print machine. We plan to have 4 belts for 2 screen printed machines. (this means we have 1 belts on each cavity, so for 4 cavities we have 4 belts)
  - Each belt needs will have different length for dropping the layers in the Feed zone, so the belt length should be adjustable. Please see the diagram below.

- Roller Coating Machine (NBR)
  - We have 1 Roller Coating Machine (NBR). We have a normal conveyor belt for feeding the layers. 2 layers are fed per stroke of the machine. We plan to have 2 belts for NBR.
  - Belt length should be adjustable. Please see the diagram below.





- The Feed zone has a Cover on it with the above dimension.
  - There are 3 openings on one side and one opening on the other side for SPT 1 machine.
  - The opening is as follow:
    - For NBR: 1285 mm
    - For SPT 1: 1545 mm
    - For SPT 2: 1295 mm
  - Height of Each Opening is 400 mm maximum

- Dimension of the layers to be transferred from SPT and NBR are as follow:

	Min	Max
Length	100mm	600mm
Width	100 mm	400mm
Height	0.1 mm	1.0 mm
Weight	14 gram	200 gram

Process flow

- The operator scan the bar code from the production order paper at the main curing oven HMI panel. The signal goes at first to the oven main panel.
- In the main panel there are predefined program which will be provided by oven supplier according to OT requirements. As the signal goes to main oven panel the curing oven start to reach the required parameters.
- Once the parameter has reached the tolerance the oven will send a signal to the transfer mechanism (which belts to run), accordingly the belt or belts will start running.
- The operator can now start his production on the screen print machine.

2. Transfer Mechanism

Mechanical Specification

Belts	Belt 1	Belt 2	Belt 3	Belt 4	Belt 5	Belt 6
	NBR		SPT-1		SPT-2	
Length(mm) (Minimum)	1100	1100	750	750	650	650
Length(mm) (Maximum)	1700	1700	1700	1700	1300	1300
Length(mm) (Fixed)	1000	1000	650	650	650	650
Width (mm)	330	330	650	650	650	650
Height (mm) (Adjustable) (from ground to the top of the belt)	950-1100	950-1100	950-1100	950-1100	950-1100	950-1100
Belt Material	PVC	PVC	PVC	PVC	PVC	PVC
Belt Speed (m/min)	0-10	0-10	0-10	0-10	0-10	0-10
Idler Diameter for the belt (mm)	50	50	50	50	50	50

Technical Parameters

- Belt Conveyor:
  - Makes no imprints or scratches on the layers
  - Belt material should be PVC
  - Belt durability
  - Easy to change
  - Belt tension should be maintained all the time
  - Belt aligning must be provided
- Motor should be servomotor. NOTE: The motor will be near explosive area. (Suitable option for the same can be used with prior approval from OT)
- Gap between belts
  - Belt 1 & Belt 2 – 40 mm
  - Belt 3 & Belt 4 – 40 mm
  - Belt 5 & Belt 6 – 40 mm
- The whole Transfer Mechanism system will rest on floor with resting pads and wheels are also need to be provided for easy movement.
- The conveyor belts should be open.(Belt case covering)
- Conveyor orientation should be flat throughout the length.
- The construction needs to be built in such a way that no big movements will be happen during using.
- To ensure the correct positioning each unit could be fix with a physically structure with the feed zone
- No vibration has to be there at free end of all conveyors. If required support can be given from other side with prior approval from EK)
- Aluminum profiles to be used for structure.
- There should be no deflection in angle of the layers while transferring from machine end to Feed zone belt. If required additional structure can be added in feed zone to achieve zero deflection.

Electrical Parameters

- The Transfer mechanism will be connected with the main oven line for communication between Transfer mechanism and Oven
- The Connection Interface will be PROFInet
- PLC Connection type
  - Siemens 7-1515F-2PN
- PN/PN coupler coupling module for connecting two PROFInet network
  - Model No. 6ES7158-3ADO1-OXAO
  - Mounting: mounting rail 7.5mm and 15mm
  - Quantity -1
- HMI Panel
  - One HMI Panel and emergency stop button for all the Belts

3. Additional line information

- Spare part package (optional)
- All local and country specification must be follow
- Supplier logo is defend to put on the machine
- Oven colour RAL 7035 (light gray)
- Quote should have DAP price and ex works price.
- Final length version will be decide during the project.

4. Energy

- According to the specific local rules and OT specification D
- Power connection (water air pressure & electric) according to the OT specification D
- Power connection should be in one area.
- All local rules and country specification must be follow (India)
- In case of using an electrical cabinet an UPS backup has to be installed cause of India power failure (power bank again power failure to avoid electrical damage)

5. Training

- Maintenance Training required
- Worker training required

6. Documentation

- Country safety standards local laws must be follow
- Complete documentation including user manual, electrical circuit diagram & function diagrams, floor plans, Wear & Spare Parts List: 1x paper, 2xCD-ROM.
- Complete Documentation according to the OT specification D
- Maintenance plan & maintenance contract according to OT order
- Information about the surface stress and the footprint of the machine and a breakdown of dynamic and static load calculation / components

7. Acceptance

- Acceptance contents must be discussed between supplier and OT.
- Pre acceptance must be made with parts by the supplier
- During the Pre acceptance OT and the supplier inscribe all the fault and defect on the machine. (important for the final approval)
- Between Pre acceptance and final acceptance the supplier must solve all the defects and fault which both has inscribe by the Pre acceptance.
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8. Further valid documents

- Specification part A
- Industrial Engineering specification

9. Delivery address

Ranjangaon Ind. Area
Tal. Shirur, District Pune 412 220
India